

Roberto Vicario

AI Software Engineer

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 LinkedIn  GitHub  Website



Summary: AI Software Engineer experienced in designing, developing, and deploying real-world AI systems for aerospace and industrial applications. Experienced across the entire AI software lifecycle, transforming complex data into scalable AI software solutions.

Technologies: Java, Python, JavaScript, Spring Boot, FastAPI, SQL, Redis, MongoDB, Git, Docker, Kubernetes, Google Cloud Platform, BigQuery, Pandas, scikit-learn, TensorFlow, OpenCV, MLflow, Airflow

Experience

AI Software Engineer Motus ml

Oct 2024 – Mar 2026
Milan, Italy

- Developed **machine learning and data-driven software solutions** for industrial applications in the **Space Economy** and Advanced Manufacturing sectors.
- Contributed to **3 industrial projects**, progressing from full-stack software development to data science and end-to-end AI engineering.
- Implemented **Python-based REST APIs (Flask)** and **Dockerized microservices** to build production-ready software systems integrating **ETL pipelines** and **AI models** for customer on-premise environments.
- Designed and developed an AI-driven defect prediction system for **ESA-regulated** manufacturing environments, generating **thousands of component-level quality predictions** across hundreds of PCB assemblies as part of my industrial **Master's Thesis** in collaboration with the R&D team.
- Deployed the solution in customer environments, enabling predictive quality assessment before production and reducing **defect analysis time from months to days**.

Data Scientist Intern University of Insubria, DiSTA Department

Oct 2023 – Jan 2024
Varese, Italy

- Conducted a machine learning research project in the **Biomedical** domain by investigating unsupervised approaches for **stress detection** using **HRV** data.
- Gained experience in formulating **research questions**, reviewing the state of the art and related **literature**, and **reproducing** published experiments.
- The project served as the foundation of my research **Bachelor's Thesis**, conducted under the supervision of a university researcher.

Education

Master of Science in Computer Science University of Insubria

Oct 2024 – Dec 2026
Varese, Italy

- **Thesis:** Development of an Industrial AI-Based Software for Risk Estimation of PCBs in Space Manufacturing

Bachelor of Science in Computer Science University of Insubria

Oct 2020 – Feb 2024
Varese, Italy

- **Thesis:** Apply Machine Learning for Stress Detection in Office Work Environments



	Listening	Speaking	Reading	Writing
Italian	C2	C2	C2	C2
English	C1	C1	C1	C1
French	B1	B1	B1	B1
Spanish	A2	A2	A2	A2

Projects

LarioNow

Ongoing

A **real-time weather nowcasting** application for the Lake Como area. Designed and implemented an automated **ETL pipeline** using **Google Cloud Run**, **Cloud Scheduler**, and **BigQuery** to collect and process meteorological data from **19 weather stations every 5 minutes**. Developed and deployed a **FastAPI** application powered by an **AutoML XGBoost** model to provide real-time, location-based **weather forecasts** for the next 30, 60, 90, and 120 minutes.

2UML

Repository 

A web application that automatically **transforms natural language** software requirements into **UML diagrams**. Developed and deployed a **FastAPI** application that leverages an **LLM** to generate editable **PlantUML** code, allowing users to refine the generated diagrams through a **built-in code editor**.

Programs

CyberChallenge.IT

Feb 2025 – Jun 2025

Cybersecurity National Lab

Varese, Italy

Admitted to the **Italian training** program as one of the **top 1000 U24 ethical hackers nationwide**. The training featured hands-on challenges in cryptography, penetration testing (**pwntools**), network traffic analysis (**Wire-shark**), web application security (**SQLi**, **XSS**), binary exploitation (**Ghidra**), code injection, and intelligence operations.

Certifications and Licenses

CyberChallenge.IT

Jun 2025

Cybersecurity National Lab – PDF

Cybersecurity Attack and Defense Fundamentals

Oct 2023

EC-Council – PDF

Driving License

Jun 2020

Type: B · Car Owner: Yes